

Open source software in the enterprise

Operating system

The most popular choice for an open source operating system is Linux, due to its vast community support and adoption.

There are multiple Linux distributions for enterprises, which vary in the packages bundled and the certified applications that accompany it. Red Hat Enterprise Linux (RHEL), SUSE Linux Enterprise Server (SLES), Oracle Linux and Ubuntu Enterprise Linux are a few.

Linux is also the favourite OS due to its support for container deployment features.

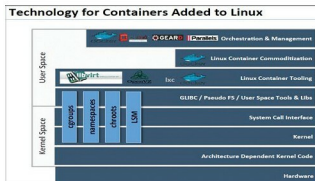


Figure 4: Linux technology for supporting containers (Source: <http://image.slide-sharecdn.com/dockeroverview-riseofthecontainers-150417102414-conversion-gate02/95/docker-overview-rise-of-the-containers-7-638.jpg?cb=1429267165>)

Virtualisation software

The virtualisation method can be categorised based on how it mimics hardware to a guest operating system and emulates the guest operating environment. Primarily, there are three types of virtualisation:

- Emulation
- Para-virtualisation
- Container-based virtualisation

Emulation: Emulation, also known as full virtualisation, runs the virtual machine OS kernel entirely in software. The hypervisor used in this type is known as a Type 2 hypervisor. It is installed on top of the host operating system, which is responsible for translating the guest OS kernel code to software instructions. The translation is done entirely in software and requires no hardware involvement. Emulation makes it possible to run any non-modified operating system that supports the environment being emulated. The downside of this type of virtualisation is the additional system resource overhead that leads to a decrease in performance compared to other types of virtualisation.

Oracle VM VirtualBox and QEMU are the top choices for the emulation kind of virtualisation.

QEMU is a hosted virtual machine monitor. It emulates CPUs through dynamic binary translation and provides a set of device models, enabling it to run a variety of unmodified guest operating systems. It can also be used with KVM to run virtual

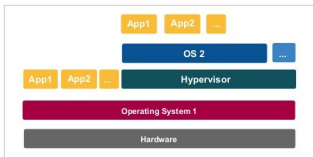


Figure 5: Type 2 hypervisor

machines at near-native speed (requiring hardware virtualisation extensions on x86 machines). QEMU can also be used purely for CPU emulation for user-level processes, allowing applications compiled for one architecture to be run on another.

Oracle VM VirtualBox (formerly Sun VirtualBox) is a free and open source hypervisor for x86 computers currently being developed by Oracle Corporation. VirtualBox may be installed on a number of host operating systems, including Linux, OS X, Windows, Solaris and Open Solaris. There are also ports to FreeBSD and Genode. It also supports the creation and management of guest virtual machines that run versions and derivations of Windows, Linux, BSD, OS/2, Solaris, Haiku, OSx86 and limited virtualisation of OS X guests on Apple hardware.

Para-virtualisation: It is also known as a Type 1 hypervisor, running directly on the hardware or 'bare-metal', and provides virtualisation services directly to the virtual machines running on it. It helps the operating system, the virtualised hardware, and the real hardware to collaborate to achieve optimal performance. These hypervisors typically have a rather small footprint and do not, themselves, require extensive resources.

Featured below are the top two open source para-virtualisation products.

KVM (Kernel-based Virtual Machine) is a full virtualisation solution for Linux on x86 hardware containing virtualisation extensions (Intel VT or AMD-V). It consists of a loadable kernel module, *kvm.ko*, that provides the core virtualisation infrastructure and a processor-specific module like *kvm-intel.ko* or *kvm-amd.ko*.

Using KVM, one can run multiple virtual machines running unmodified Linux or Windows images. Each virtual machine has private virtualised hardware—a network card,

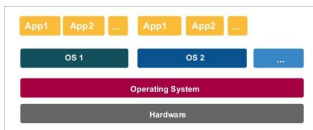


Figure 6: Para-virtualisation